

ARCHITOP®

High resistance concrete-effect flooring with minimal thickness
(3 mm)

DESCRIPTION

ARCHITOP® is an about 3 mm thick concrete-effect flooring characterized by high strength and exceptional aesthetic qualities. It consists of a water-based copolymer liquid mixed with IDEAL WORK COLOUR-HARDENER colored with pigments from the COLOR PACK-C range. The surface is subsequently smoothed with a power trowel until a smooth, closed finish is achieved.

FEATURES

Architop® can be applied on new solid horizontal surfaces or over existing floors, has an extremely low thickness (3 mm) and a practical and easy application processing. Architop® offers the aesthetic effect of concrete flooring without the drawbacks that mass concrete installation entails. It is available in a wide range of colours. Architop® does not require expansion joints although existing joints on the substrate must be respected.

FIELDS OF USE

- External or internal horizontal surfaces where high resistance in low thickness is required.
- Overcoating old ceramic or marble flooring.
- In renovations where there are load and/or height restrictions.
- As an alternative to the classic industrial floor for shops and offices where a more refined finish and colour scheme is desired.
- As an alternative to cast concrete flooring, in construction site situations, where faster, less disruptive work is required.
- For horizontal surfaces subject to high traffic

APPLICATION

SURFACE PREPARATION

This depends on the type and condition of the substrate (concrete, sand-cement, ceramic, self-leveling...).

The concrete substrate must have a compressive strength of at least 25 Mpa and 1.0 Mpa at pull-off and a residual humidity of not more than 4.0%.

The surface must be pretreated with sanding, shot blasting or high pressure washing and must also be clean and free from contamination, be rough and porous.

On the prepared surface, apply and spread at zero EPOXY-COAT NV mixed with up to 10% of quartz 0.1-0.5 mm and possibly diluted up to 5% with X100 solvent. Fully broadcast with quartz 0.7-1.2 mm until saturation (about 2.5 Kg/m²).

Smooth with roller to remove trowel marks before broadcasting. Limit the work area so that broadcasting can be completed while the resin is still wet.

The entire surface must be completely covered with quartz. Always wear spiked shoes or cleats to walk on the surface.

Preparation on a ceramic background

1. Always sand with diamond disc.
2. Apply a fibreglass mesh fixed with IW-BLOCKER
3. On a base with evident joints, apply 2 coats of epoxy-coat with quartz broadcast (see above), on ceramic without or with minimal joints, just 1 coat is enough.

Preparation on concrete in the absence of rising damp

1. Sand or shotblast.
2. Apply a coat of epoxy-coat with quartz broadcast (see above).

Preparation on concrete in the presence of rising damp

1. Sand or shotblast.
2. Apply a coat of BARRIERA CEM
3. Apply a coat of epoxy-coat with quartz broadcast (see above).

Preparation on sand and cement base

1. Sand with paper 24
2. Possible consolidation (if necessary) with IDEAL WATER
3. Possible consolidation (if necessary) with IW-BLOCKER and fiberglass mesh (alternative)
4. Apply a coat of epoxy-coat with quartz broadcast (see above)

Self-leveling preparation

1. Grind
2. Consolidate with IDEAL WATER
3. Apply a coat of epoxy-coat with quartz broadcast (see above).

APPLICATION OF THE ARCHITOP® MIXTURE

Composition of the mixture for the first coat (quantity for about 12.5 m²) Color Hardener 25 Kg + Architop Catalyst pure 6 Kg

Composition of the mixture for the second coat (quantity for about 15.0 m²) Color Hardener 25 Kg + Architop Catalyst 4 Kg with 2 L of water.

Mix Architop Catalyst, COLOR PACK-C and ARCHI-GO and thoroughly for at least 2 minutes, then mix it with COLOUR-HARDENER.

Spread the mixture evenly with a spatula or squeegee and remove any lumps.

Apply approximately 3.5 Kg of the mixture in 2 coats (2 Kg the first coat, 1.5 Kg second). This creates a thickness of about 3 mm.

The second coat must be applied when the first is still wet but is not tacky to the touch and does not allow the movement of the aggregate. Premature application of the second coat would move the already laid material, while a delayed laying would affect its adhesion.

POWER TROWEL

Use machines with 60 -75 -90 diameter, which can also be operated in limited environments.

Combustion or electric machines must have a power of at least 2.2 kW. Make sure that the blades

are intact and sharp, if necessary, sharpen them with a flex tool.

Start the first pass and then the following only when the mixture will appear compact and moist but the material will not stick on touch. Touching must not move the material. Check consistency with your hand. The interval between one pass and the other will be longer at lower temperatures and with higher humidity and shorter in the reverse conditions.

The first two steps are carried out with the disc, the following with the blades which will gradually be used more inclined as the surface dries. The final step will be carried out almost at maximum inclination. For complete closure of the surface, 7-9 passes with the blades are necessary. Make cross passages. For light-colored surfaces, it is recommended to perform the initial finishing passes using Teflon blades to minimize the risk of burn marks. The final 2 to 3 passes can then be completed using traditional steel blades.

By stopping the processing after the first step, the surface remains rough, suitable for outdoor use whilst with the subsequent steps it acquires the characteristic smooth, closed and cloudy structure. The perimeter bands along the walls must be finished by hand trowel after passing with the power trowel. Always keep the wall on the left and proceed so that the blades, which rotate clockwise, push the material towards the wall itself.

Usually 3 passes by hand trowel are sufficient on the perimeter bands: the first pass by hand trowel is after the first pass of the disc, the second after the first pass with the blades, the third after one of the last passes when the surface starts to close and acquire the cloudy aspect.

During the previous processing step use IW-EC in order to soften the surface and create a kind of soft grout that can be easily smoothed with the trowel. IW-EC must also be used in areas where the material tends to harden, increasing friction with the blades.

Once the perimeter area has been processed, remove with the trowel the possible left marks.

Remain 10-20 cm from the wall, taking care not to damage the smoothed part.

Use spiked shoes or cleats to walk on the fresh floor during the second step of coating and during the first trowel passes. For the finishing phase, wear smooth footwear, preferably with a soft sponge layer under the sole.

Keep tools clean whilst working. The helicopter blades, must be cleaned after each pass.

SURFACE PROTECTION

Once curing has been fully completed, generally not earlier than 24-36 hours after application, one of the following treatments may be carried out:

- Water- and oil-repellent impregnating protective treatment with PETROTEX-S;
- Crystallizing treatment with IDEAL-HARD-WR;
- Epoxy-polyurethane treatment consisting of two coats of IDEALPU-PRIMER MAXEP and one or two (recommended) coats of IDEALPU-WB EASY.

Please refer to the relevant Technical Data Sheets for further details.

TECHNICAL DATA

See the safety data-sheet <https://www.idealwork.it/download/documentazione-tecnica>

CONSUMPTION

1st Coat:

HARD-NEU	COLOUR-HARDENER	2,00 Kg/m ²
ARCHI-CATA	ARCHITOP® CATALYST	0,480 Kg/m ²
ARCHI-GO		0,040 Kg/m ²

2nd coat

HARD-NEU	COLOUR-HARDENER	1,50 Kg/m ²
ARCHI-CATA	ARCHITOP® CATALYST	0,240 Kg/m ²
ARCHI-GO		0,020 Kg/m ²

IW-EC		7,5-10 g/m ²
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PACKAGING/STORAGE/DISPOSAL

EPOXY-COAT NV	Kg 22,60
COLOUR-HARDENER	Kg 25,00
ARCHITOP CATALYST	Kg 25,00
ARCHI-GO	Kg 10,00

Consult the data sheets for the relevant products

PRECAUTIONS

Avoid conditions that may encourage differential setting times between the areas, such as air currents and solar radiation. Any underfloor heating must be turned off 5 days before the ARCHITOP® installation.

The separation between two ARCHITOP® applications is carried out with the specific blue 5 cm canvas tape. This protects the adjacent part and creates a precise guide for the subsequent cutting of the separation joint. The tape must be applied firstly before preparation with Epoxy-coat, then before applying the Architop. It will be removed before sanding once the material is completely dry. Any shrinkage cracks that formed in the substrate inevitably tend to be transmitted to the ARCHITOP® finishing layer. If the substrate is characterized by a 28 days shrinkage of more than 300 µm / m (UNI 11307 for concrete and UNI 6687 for substrates in sand and cement or mortar) it is advisable to wait for it to fully cure and seal any cracks with Epoxy-coat before ARCHITOP® application.

IMPORTANT:

The manufacturer disclaims any responsibility for the achievement of the services declared for the system and for the success of the work if it is not used in accord with its instructions. The manufacturer disclaims any responsibility for the achievement of the services declared for the system and for the success of the work if it is performed even partially in contrast with these guidelines or with products not covered by them.

The manufacturer also declines any responsibility for the aesthetic aspect of the flooring which depends on the installation methods, working times and climatic conditions of the site. The applicator must take these into account when scheduling and applying the materials supplied by the manufacturer.

The applicator notes that Ideal Work is in no way responsible for the suitability of the solution chosen in relation to:

A) Substrate suitability, climatic conditions or any other external parameter which may affect the performance of the Ideal Work products being used.

B) The stresses to which Ideal Work products may be put under operation.

It also notes that the indications provided by Ideal Work in its technical documentation are to be considered a necessary condition but do not in any way relieve the contractor of the responsibilities and technical evaluations of the applicator. The data can be changed at any time. Also note that the products are intended for professional use only.